Class: **CAPE Pure Mathematics Unit 1**

Topic: **Past Paper June 2014 – Paper 2 – Solutions**

June 2014 – Question 1 – Solution

(a) Let $\boldsymbol{p}, \boldsymbol{q}$ and $\boldsymbol{r}$ be three propositions. Construct a truth table for the statement

$$(\boldsymbol{p} \to \boldsymbol{q}) \wedge (\boldsymbol{r} \to \boldsymbol{q})$$ [5]

The truth table for the statement $(\boldsymbol{p} \to \boldsymbol{q}) \wedge (\boldsymbol{r} \to \boldsymbol{q})$ is:

$\boldsymbol{p}$	$\boldsymbol{q}$	$\boldsymbol{r}$	$\boldsymbol{p} \to \boldsymbol{q}$	$\boldsymbol{r} \to \boldsymbol{q}$	$(\boldsymbol{p} \to \boldsymbol{q}) \wedge (\boldsymbol{r} \to \boldsymbol{q})$
T	T	T	T	T	T
T	T	F	T	T	T
T	F	T	F	F	F
T	F	F	F	T	F
F	T	T	T	T	T
F	T	F	T	T	T
F	F	T	T	F	F
F	F	F	T	T	T

(b) A binary operator $\oplus$ is defined on a set of positive real numbers by

$$y \oplus x = y^3 + x^3 + ay^2 + ax^2 - 5y - 5x + 16 \text{ where } a \text{ is a real number.}$$

(i) State, giving a reason for your answer, if $\oplus$ is commutative in $\mathbb{R}$. [3]

$\oplus$ is commutative in $\mathbb{R}$ if $y \oplus x = x \oplus y$.

$$y \oplus x = y^3 + x^3 + ay^2 + ax^2 - 5y - 5x + 16$$

$$= x^3 + y^3 + ax^2 + ay^2 - 5x - 5y + 16$$

$$= x \oplus y$$

Since $y \oplus x = x \oplus y$, then $\oplus$ is commutative in $\mathbb{R}$.

(ii) Given that $f(x) = 2 \oplus x$ and $(x - 1)$ is a factor of $f(x)$,

(a) find the value of a [4]

$$f(x) = 2 \oplus x$$

$$= (2)^3 + x^3 + a(2)^2 + ax^2 - 5(2) - 5xy + 16$$

$$= 8 + x^3 + 4a + ax^2 - 10 - 5x + 16$$

$$= x^3 + ax^2 - 5x + 4a + 14$$

Since $(x - 1)$ is a factor of $f(x)$, then by the Factor Theorem, $f(1) = 0$.

$$f(1) = (1)^3 + a(1)^2 - 5(1) + 4a + 14$$

$$0 = 1 + a - 5 + 4a + 14$$

$$0 = 5a + 10$$

$$5a = -10$$

$$a = \frac{-10}{5}$$

$$a = -2$$

$\therefore$ The value of $a = -2$.

(b) factorize $f(x)$ completely [3]

$$f(x) = x^3 + (-2)x^2 - 5x + 4(-2) + 14$$

$$= x^3 - 2x^2 - 5x - 8 + 14$$

$$= x^3 - 2x^2 - 5x + 6$$

We are given that $(x - 1)$ is a factor of $f(x)$.

By long division,

$$\begin{array}{r|l} & x^2 - x - 6 \\ \hline x - 1 & x^3 - 2x^2 - 5x + 6 \\ & x^3 - x^2 \\ \hline & -x^2 - 5x \\ & -x^2 + x \\ \hline & -6x + 6 \\ & -6x + 6 \\ \hline & 0 \end{array}$$

So, we have,

$$f(x) = (x - 1)(x^2 - x - 6)$$

$$= (x - 1)(x - 3)(x + 2)$$

(c) Use mathematical induction to prove that

$$1^2 + 3^2 + 5^2 + \cdots + (2n-1)^2 = \frac{n}{3}(4n^2 - 1) \text{ for } n \in \mathbb{N}$$ [10]

Let $P(n)$ be the statement: $1^2 + 3^2 + 5^2 + \cdots + (2n-1)^2 = \frac{n}{3}(4n^2 - 1)$ for $n \in \mathbb{N}$.

Looking at $P(1)$:

L.H.S. $= 1^2$

$= 1$

R.H.S. $= \frac{1}{3}[4(1)^2 - 1]$

$= \frac{1}{3}(4 - 1)$

$= \frac{1}{3}(3)$

$= 1$

Since L.H.S. = R.H.S., $P(1)$ is true.

Assume $P(k)$ is true.

That is, $1^2 + 3^2 + 5^2 + \cdots + (2k-1)^2 = \frac{k}{3}(4k^2 - 1)$

Looking at $P(k+1)$.

That is, $1^2 + 3^2 + 5^2 + \cdots + [2(k+1) - 1]^2 = \frac{k+1}{3}[4(k+1)^2 - 1]$

$$1^2 + 3^2 + 5^2 + \cdots + (2k+1)^2 = \frac{k+1}{3}[4(k+1)^2 - 1]$$

We need to prove $P(k+1)$.

Taking L.H.S:

$$1^2+3^2+5^2+\cdots+(2k+1)^2 = 1^2+3^2+5^2+\cdots+(2k-1)^2+(2k+1)^2$$

$$= \frac{k}{3}(4k^2-1)+(2k+1)^2 \qquad \text{[from assumption]}$$

$$= \frac{k}{3}(2k+1)(2k-1)+(2k+1)(2k+1)$$

$$= \frac{2k+1}{3}[k(2k-1)+3(2k+1)]$$

$$= \frac{2k+1}{3}(2k^2-k+6k+3)$$

$$= \frac{2k+1}{3}(2k^2+5k+3)$$

$$= \frac{2k+1}{3}\cdot(k+1)(2k+3)$$

$$= \frac{k+1}{3}\cdot(2k+1)(2k+3)$$

$$= \frac{k+1}{3}\cdot(4k^2+8k+3)$$

$$= \frac{k+1}{3}\cdot(4k^2+8k+4-1)$$

$$= \frac{k+1}{3}[4(k^2+2k+1)-1]$$

$$= \frac{k+1}{3}[4(k+1)^2-1]$$

$$= \text{R.H.S.}$$

$\therefore$ If $P(k)$ is true, then $P(k+1)$ is also true.

Since $P(1)$ is true, and since $P(k+1)$ is true when $P(k)$ is true, then by the principle of mathematical induction, $P(n)$ is true for all n, ie.

$1^2+3^2+5^2+\cdots+(2n-1)^2 = \frac{n}{3}(4n^2-1)$ for $n \in \mathbb{N}$. Q.E.D.

Total: 25 marks

June 2014 – Question 2 – Solution

(a) The functions f and g are defined as follows:

$$f(x) = 2x^2 + 1$$

$$g(x) = \sqrt{\frac{x-1}{2}}$$

where $1 \leq x \leq \infty, x \in \mathbb{R}$.

(i) Determine, in terms of x,

(a) $f^2(x)$ [3]

$$\begin{aligned} f^2(x) &= f[f(x)] \\ &= f(2x^2 + 1) \\ &= 2(2x^2 + 1)^2 + 1 \\ &= 2(4x^4 + 4x^2 + 1) + 1 \\ &= 8x^4 + 8x^2 + 2 + 1 \\ &= 8x^4 + 8x^2 + 3 \end{aligned}$$

(b) $f[g(x)]$ [3]

$$\begin{aligned} f[g(x)] &= f\left(\sqrt{\frac{x-1}{2}}\right) \\ &= 2\left(\sqrt{\frac{x-1}{2}}\right)^2 + 1 \\ &= 2\left(\frac{x-1}{2}\right) + 1 \\ &= x - 1 + 1 \\ &= x \end{aligned}$$

(ii) Hence, or otherwise, state the relationship between f and g. [1]

Since $f[g(x)] = x$, then $g(x) = f^{-1}(x)$.

The relationship is that g is the inverse function of f.

(b) Given that $a^3 + b^3 + 3a^2b = 5ab^2$, show that $3\log\left(\frac{a+b}{2}\right) = \log a + 2\log b$. [5]

Required to prove that $3\log\left(\frac{a+b}{2}\right) = \log a + 2\log b$.

Taking L.H.S:

$$3\log\left(\frac{a+b}{2}\right) = \log\left(\frac{a+b}{2}\right)^3$$

$$= \log\left(\frac{a^3+3a^2b+3ab^2+b^3}{8}\right)$$

$$= \log\left(\frac{5ab^2+3ab^2}{8}\right)$$

$$= \log\left(\frac{8ab^2}{8}\right)$$

$$= \log(ab^2)$$

$$= \log a + \log b^2$$

$$= \log a + 2\log b$$

$$= \text{R.H.S.}$$

$$\therefore 3\log\left(\frac{a+b}{2}\right) = \log a + 2\log b$$

Q.E.D.

(c) Solve EACH of the following equations:

(i) $e^x + \frac{1}{e^x} - 2 = 0$ [4]

$e^x + \frac{1}{e^x} - 2 = 0$

$(\times e^x)$

$e^{2x} + 1 - 2e^x = 0$

$e^{2x} - 2e^x + 1 = 0$

$(e^x - 1)(e^x - 1) = 0$

$(e^x - 1)^2 = 0$

$e^x - 1 = 0$

$e^x = 1$

Taking ln on both sides:

$\ln e^x = \ln 1$

$x \ln e = 0$

$x = 0$

(ii) $\log_2(x+1) - \log_2(3x+1) = 2$ [4]

$\log_2(x+1) - \log_2(3x+1) = 2$

$\log_2\left(\frac{x+1}{3x+1}\right) = 2$

Converting to exponential form gives:

$\frac{x+1}{3x+1} = 2^2$

$\frac{x+1}{3x+1} = 4$

$$x + 1 = 4(3x + 1)$$

$$x + 1 = 12x + 4$$

$$x - 12x = 4 - 1$$

$$-11x = 3$$

$$x = -\frac{3}{11}$$

(d) Without the use of a calculator, show that

$$\frac{\sqrt{3}-1}{\sqrt{3}+1} + \frac{\sqrt{3}+1}{\sqrt{3}-1} + \frac{\sqrt{2}-1}{\sqrt{2}+1} + \frac{\sqrt{2}+1}{\sqrt{2}-1} = 10 \qquad [5]$$

Taking L.H.S:

$$\frac{\sqrt{3}-1}{\sqrt{3}+1} + \frac{\sqrt{3}+1}{\sqrt{3}-1} + \frac{\sqrt{2}-1}{\sqrt{2}+1} + \frac{\sqrt{2}+1}{\sqrt{2}-1} = \frac{\sqrt{3}-1}{\sqrt{3}+1} \cdot \frac{\sqrt{3}-1}{\sqrt{3}-1} + \frac{\sqrt{3}+1}{\sqrt{3}-1} \cdot \frac{\sqrt{3}+1}{\sqrt{3}+1} + \frac{\sqrt{2}-1}{\sqrt{2}+1} \cdot \frac{\sqrt{2}-1}{\sqrt{2}-1} + \frac{\sqrt{2}+1}{\sqrt{2}-1} \cdot \frac{\sqrt{2}+1}{\sqrt{2}+1}$$

$$= \frac{3-2\sqrt{3}+1}{3-1} + \frac{3+2\sqrt{3}+1}{3-1} + \frac{2-2\sqrt{2}+1}{2-1} + \frac{2+2\sqrt{2}+1}{2-1}$$

$$= \frac{3-2\sqrt{3}+1}{2} + \frac{3+2\sqrt{3}+1}{2} + \frac{2-2\sqrt{2}+1}{1} + \frac{2+2\sqrt{2}+1}{1}$$

$$= \frac{3-2\sqrt{3}+1+3+2\sqrt{3}+1}{2} + \frac{2-2\sqrt{2}+1+2+2\sqrt{2}+1}{1}$$

$$= \frac{3+1+3+1}{2} + \frac{2+1+2+1}{1}$$

$$= \frac{8}{2} + 6$$

$$= 10$$

$$= \text{R.H.S.}$$

$$\therefore \frac{\sqrt{3}-1}{\sqrt{3}+1} + \frac{\sqrt{3}+1}{\sqrt{3}-1} + \frac{\sqrt{2}-1}{\sqrt{2}+1} + \frac{\sqrt{2}+1}{\sqrt{2}-1} = 10$$

Q.E.D.

Total: 25 marks

June 2014 – Question 3 – Solution

(a) (i) Prove that $\frac{\cot y-\cot x}{\cot x+\cot y}=\frac{\sin(x-y)}{\sin(x+y)}$. [4]

Required to prove that $\frac{\cot y-\cot x}{\cot x+\cot y}=\frac{\sin(x-y)}{\sin(x+y)}$.

Taking L.H.S:

$$\frac{\cot y-\cot x}{\cot x+\cot y}=\frac{\frac{\cos y}{\sin y}-\frac{\cos x}{\sin x}}{\frac{\cos x}{\sin x}+\frac{\cos y}{\sin y}}$$

$$=\left(\frac{\cos y}{\sin y}-\frac{\cos x}{\sin x}\right)\div\left(\frac{\cos x}{\sin x}+\frac{\cos y}{\sin y}\right)$$

$$=\frac{\sin x\cos y-\cos x\sin y}{\sin x\sin y}\div\frac{\cos x\sin y+\sin x\cos y}{\sin x\sin y}$$

$$=\frac{\sin x\cos y-\cos x\sin y}{\sin x\sin y}\times\frac{\sin x\sin y}{\cos x\sin y+\sin x\cos y}$$

$$=\frac{\sin x\cos y-\cos x\sin y}{\cos x\sin y+\sin x\cos y}$$

$$=\frac{\sin(x-y)}{\sin(x+y)}$$

= R.H.S.

$$\therefore\ \frac{\cot y-\cot x}{\cot x+\cot y}=\frac{\sin(x-y)}{\sin(x+y)}$$

Q.E.D.

(ii) Hence, or otherwise, find the possible values for y in the trigonometric equation

$$\frac{\cot y - \cot x}{\cot x + \cot y} = 1, \quad 0 \le y \le 2\pi,$$

when $\sin x = \frac{1}{2}, 0 \le x \le \frac{\pi}{2}$. [8]

$$\frac{\cot y - \cot x}{\cot x + \cot y} = 1$$

$$\frac{\sin(x-y)}{\sin(x+y)} = 1$$

$$\sin(x-y) = \sin(x+y)$$

$$\sin x \cos y - \sin y \cos x = \sin x \cos y + \sin y \cos x$$

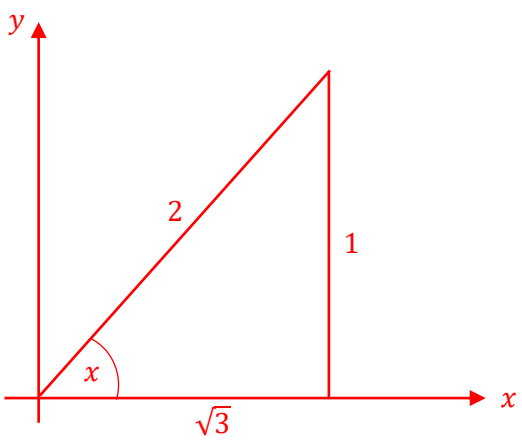
$$-\sin y \cos x = \sin y \cos x$$

$$0 = \sin y \cos x + \sin y \cos x$$

$$2 \sin y \cos x = 0$$

We are given that $\sin x = \frac{1}{2}$

$$= \frac{opp}{adj}$$



By Pythagoras' Theorem,

$$a^2 + b^2 = c^2$$

$$a^2 + (1)^2 = (2)^2$$

$$a^2 + 1 = 4$$

$$a^2 = 4 - 1$$

$$a^2 = 3$$

$$a = \sqrt{3}$$

Now,

$$\cos x = \frac{adj}{hyp}$$

$$= \frac{\sqrt{3}}{2}$$

Substituting $\cos x = \frac{\sqrt{3}}{2}$ into $2 \sin y \cos x = 0$ gives:

$$2 \sin y \left(\frac{\sqrt{3}}{2}\right) = 0$$

$$\sqrt{3} \sin y = 0$$

$$\sin y = 0$$

$$y = 0, \pi, 2\pi$$

$\therefore$ The possible values of y are $y = 0, \pi, 2\pi$ for the range $0 \leq y \leq 2\pi$.

(b) (i) Express $f(\theta) = 3\sin 2\theta + 4\cos 2\theta$ in the form $r\sin(2\theta + \alpha)$ where

$r > 0$ and $0 < a < \frac{\pi}{2}$. [4]

$f(\theta) = 3\sin 2\theta + 4\cos 2\theta$

Now,

$3\sin 2\theta + 4\cos 2\theta = r\sin(2\theta + \alpha)$

$3\sin 2\theta + 4\cos 2\theta = r\sin 2\theta\cos\alpha + r\cos 2\theta\sin\alpha$

Equating coefficients of $\sin 2\theta$ and $\cos 2\theta$ gives:

$3 = r\cos\alpha$ $\rightarrow$ Equation 1

$4 = r\sin\alpha$ $\rightarrow$ Equation 2

Equation 2 ÷ Equation 1 gives:

$\frac{r\sin\alpha}{r\cos\alpha} = \frac{4}{3}$

$\tan\alpha = \frac{4}{3}$

$\alpha = \tan^{-1}\left(\frac{4}{3}\right)$

$\alpha = 0.927^c$ (to 3 significant figures)

And,

$r = \sqrt{3^2 + 4^2}$

$= \sqrt{9 + 16}$

$= \sqrt{25}$

$= 5$

$\therefore 3\sin 2\theta + 4\cos 2\theta = 5\sin(2\theta + 0.927^c)$

(ii) Hence, or otherwise, determine

(a) the value of θ, between 0 and 2π radians, at which $f(\theta)$ is a **minimum**. [4]

$f(\theta) = 3\sin 2\theta + 4\cos 2\theta$

$f(\theta) = 5\sin(2\theta + 0.927^c)$

Now,

$-1 \leq \sin\theta \leq 1$

$-1 \leq \sin(2\theta + 0.927^c) \leq 1$

$-5 \leq 5\sin(2\theta + 0.927^c) \leq 5$

So, the minimum value of $f(\theta)$ is -5.

Now,

$$5\sin(2\theta + 0.927^c) = -5$$

$$\sin(2\theta + 0.927^c) = -1$$

$$2\theta + 0.927^c = \frac{3\pi}{2}, \frac{7\pi}{2}$$

$$2\theta = 3.785^c, 10.069^c$$

$$\theta = 1.89^c, 5.03^c$$

$\therefore$ The value of θ at which $f(\theta)$ is a minimum is $\theta = 1.89^c$ and $\theta = 5.03^c$.

(b) the minimum and maximum values of $\frac{1}{7 - f(\theta)}$. [5]

$$f(\theta) = 5\sin(2\theta + 0.927^c)$$

$$-1 \leq \sin\theta \leq 1$$

$$-1 \leq \sin(2\theta + 0.927^c) \leq 1$$

$$-5 \leq -5\sin(2\theta + 0.927^c) \leq 5$$

$$7 - 5 \leq 7 - 5\sin(2\theta + 0.927^c) \leq 7 + 5$$

$$2 \leq 7 - 5\sin(2\theta + 0.927^c) \leq 12$$

$$2 \leq 7 - f(\theta) \leq 12$$

$$\frac{1}{12} \leq \frac{1}{7 - f(\theta)} \leq \frac{1}{2}$$

$\therefore$ Minimum value of $\frac{1}{7-f(\theta)}$ is $\frac{1}{12}$.

$\therefore$ Maximum value of $\frac{1}{7-f(\theta)}$ is $\frac{1}{2}$.

Total: 25 marks

June 2014 – Question 4 – Solution

(a) Let L_1 and L_2 be two diameters of a circle C. The equations of L_1 and L_2 are

$x - y + 1 = 0$ and $x + y - 5 = 0$, respectively.

(i) Show that the coordinates of the centre of the circle, C, where L_1 and L_2 intersect are $(2, 3)$. [3]

L_1: $x - y + 1 = 0$ → Equation 1

L_2: $x + y - 5 = 0$ → Equation 2

Rearranging Equation 1 gives:

$x = y - 1$ → Equation 3

Substituting Equation 3 into Equation 2 gives:

$$(y - 1) + y - 5 = 0$$
$$y + y = 5 + 1$$
$$2y = 6$$
$$y = \frac{6}{2}$$
$$y = 3$$

Substituting $y = 3$ into Equation 3 gives:

$$x = 3 - 1$$
$$= 2$$

∴ The coordinates of the centre of the circle, C, is $(2, 3)$.

Q.E.D.

(ii) A and B are endpoints of the diameter L_1. Given that the coordinates of A are $(1, 2)$ and that the diameters of a circle **bisect** each other, determine the coordinates of B. [3]

The midpoint of AB is $C(2, 3)$.

The coordinates of A are $(1, 2)$.

Let the coordinates of $B = (x_1, y_1)$.

Then, we have,

$(2, 3) = \left(\frac{x_1+1}{2}, \frac{y_1+2}{2}\right)$

Equating x and y coordinates gives:

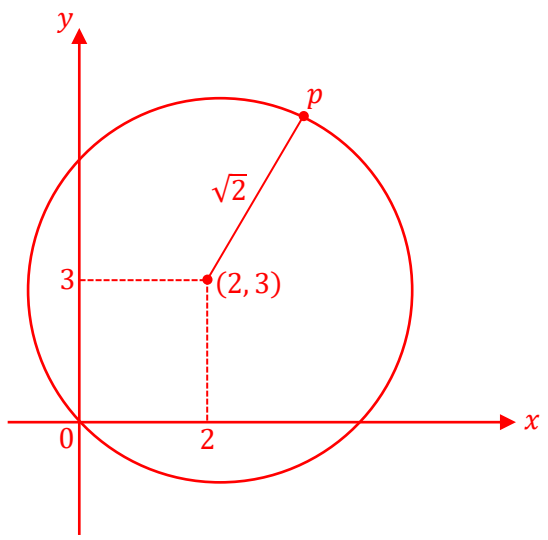
$2 = \frac{x_1+1}{2}$	and	$3 = \frac{y_1+2}{2}$
$x_1 + 1 = 4$		$y_1 + 2 = 6$
$x_1 = 4 - 1$		$y_1 = 6 - 2$
$x_1 = 3$		$y_1 = 4$

$\therefore$ The coordinates of B are $(3, 4)$.

(iii) A point, p, moves in the $x - y$ plane such that its distance from $C(2, 3)$ is always $\sqrt{2}$ units. Determine the locus of p. [3]

The locus of p is a circle with centre $(2, 3)$ and radius $\sqrt{2}$ units.

Consider the diagram below:



The equation of the locus of p is:

$(x-2)^2 + (y-3)^2 = \left(\sqrt{2}\right)^2$

$(x-2)^2 + (y-3)^2 = 2$

(b) The parametric equations of a curve, S, are given by

$x = \frac{1}{1+t}$ and $y = \frac{1}{1-t^2}$.

Determine the Cartesian equation of the curve, S. [6]

The parametric equations are:

$x = \frac{1}{1+t}$ → Equation 1

$y = \frac{1}{1-t^2}$ → Equation 2

Rearranging Equation 1 gives:

$$x(1+t) = 1$$

$$x + xt = 1$$

$$xt = 1 - x$$

$$t = \frac{1-x}{x} \qquad \rightarrow \text{Equation 3}$$

Substituting Equation 3 into Equation 2 gives:

$$y = \frac{1}{1-\left(\frac{1-x}{x}\right)^2}$$

$$= 1 \div \left[1 - \left(\frac{1-x}{x}\right)^2\right]$$

$$= 1 \div \left[1 - \frac{(1-x)^2}{x^2}\right]$$

$$= 1 \div \left[\frac{x^2-(1-x)^2}{x^2}\right]$$

$$= 1 \times \left[\frac{x^2}{x^2-(1-x)^2}\right]$$

$$= \frac{x^2}{x^2-(1-x)^2}$$

$$= \frac{x^2}{x^2-(1-2x+x^2)}$$

$$= \frac{x^2}{x^2-1+2x-x^2}$$

$$= \frac{x^2}{2x-1}$$

$\therefore$ The Cartesian equation of the curve, S, is $y = \frac{x^2}{2x-1}$.

(c) The points $P(3,-2,1)$, $Q(-1,\lambda,5)$ and $R(2,1,-4)$ are three vertices of a triangle PQR.

(i) Express EACH of the vectors $\overrightarrow{PQ}$, $\overrightarrow{QR}$ and $\overrightarrow{RP}$ in the form $x\hat{\boldsymbol{\imath}} + y\hat{\boldsymbol{\jmath}} + z\hat{\boldsymbol{k}}$. [4]

Points are $P(3,-2,1)$, $Q(-1,\lambda,5)$ and $R(2,1,-4)$.

Now,

$\overrightarrow{OP} = 3\hat{\boldsymbol{\imath}} - 2\hat{\boldsymbol{\jmath}} + \hat{\boldsymbol{k}}$

$\overrightarrow{OQ} = -\hat{\boldsymbol{\imath}} + \lambda\hat{\boldsymbol{\jmath}} + 5\hat{\boldsymbol{k}}$

$\overrightarrow{OR} = 2\hat{\boldsymbol{\imath}} + \hat{\boldsymbol{\jmath}} - 4\hat{\boldsymbol{k}}$

We have,

$$\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP}$$

$$= \begin{pmatrix} -1 \\ \lambda \\ 5 \end{pmatrix} - \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} -1-3 \\ \lambda-(-2) \\ 5-1 \end{pmatrix}$$

$$= \begin{pmatrix} -4 \\ \lambda+2 \\ 4 \end{pmatrix}$$

$$= -4\hat{\boldsymbol{\imath}} + (\lambda+2)\hat{\boldsymbol{\jmath}} + 4\hat{\boldsymbol{k}}$$

which is in the form $x\hat{\boldsymbol{\imath}} + y\hat{\boldsymbol{\jmath}} + z\hat{\boldsymbol{k}}$,

where $x = -4$, $y = \lambda + 2$ and $z = 4$.

$\overrightarrow{QR} = \overrightarrow{OR} - \overrightarrow{OQ}$

$$= \begin{pmatrix} 2 \\ 1 \\ -4 \end{pmatrix} - \begin{pmatrix} -1 \\ \lambda \\ 5 \end{pmatrix}$$

$$= \begin{pmatrix} 2-(-1) \\ 1-\lambda \\ -4-5 \end{pmatrix}$$

$$= \begin{pmatrix} 3 \\ 1-\lambda \\ -9 \end{pmatrix}$$

$$= 3\hat{\boldsymbol{\imath}} + (1-\lambda)\hat{\boldsymbol{\jmath}} - 9\hat{\boldsymbol{k}}$$

which is in the form $x\hat{\boldsymbol{\imath}} + y\hat{\boldsymbol{\jmath}} + z\hat{\boldsymbol{k}}$,

where $x = 3, y = 1 - \lambda$ and $z = -9$.

And,

$\overrightarrow{RP} = \overrightarrow{OP} - \overrightarrow{OR}$

$$= \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix} - \begin{pmatrix} 2 \\ 1 \\ -4 \end{pmatrix}$$

$$= \begin{pmatrix} 3-2 \\ -2-1 \\ 1-(-4) \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ -3 \\ 5 \end{pmatrix}$$

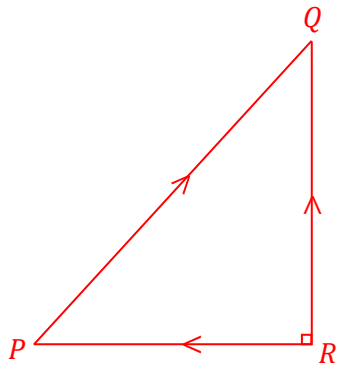
$$= \hat{\boldsymbol{\imath}} - 3\hat{\boldsymbol{\jmath}} + 5\hat{\boldsymbol{k}}$$

which is in the form $x\hat{\boldsymbol{\imath}} + y\hat{\boldsymbol{\jmath}} + z\hat{\boldsymbol{k}}$,

where $x = 1, y = -3$ and $z = 5$.

(ii) Hence, find the value of λ, given that PQR is right-angled with the side PQ as hypotenuse. [6]

Consider the diagram below:



Since $\overrightarrow{RP}$ and $\overrightarrow{RQ}$ are perpendicular to each other, then $\overrightarrow{RP} \cdot \overrightarrow{RQ} = 0$.

$$\overrightarrow{RP} \cdot \overrightarrow{RQ} = 0$$

$$\overrightarrow{RP} \cdot -\overrightarrow{QR} = 0$$

$$\begin{pmatrix} 1 \\ -3 \\ 5 \end{pmatrix} \cdot -\begin{pmatrix} 3 \\ 1-\lambda \\ -9 \end{pmatrix} = 0$$

$$\begin{pmatrix} 1 \\ -3 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ -(1-\lambda) \\ -(-9) \end{pmatrix} = 0$$

$$\begin{pmatrix} 1 \\ -3 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ -1+\lambda \\ 9 \end{pmatrix} = 0$$

$$(1)(-3) + (-3)(-1+\lambda) + (5)(9) = 0$$

$$-3 + 3 - 3\lambda + 45 = 0$$

$$-3\lambda + 45 = 0$$

$$3\lambda = 45$$

$$\lambda = \frac{45}{3}$$

$$\lambda = 15$$

Total: 25 marks

June 2014 – Question 5 – Solution

(a) Let $f(x)$ be a function defined as

$$f(x) = \begin{cases} ax + 2, & x < 3 \\ ax^2, & x \geq 3 \end{cases}$$

(i) Find the value of a if $f(x)$ is continuous at $x = 3$. [4]

Since $f(x)$ is continuous at $x = 3$, then

$$\lim_{x \to 3^-} f(x) = \lim_{x \to 3^+} f(x)$$

$$\lim_{x \to 3^-} (ax + 2) = \lim_{x \to 3^+} ax^2$$

$$a(3) + 2 = a(3)^2$$

$$3a + 2 = 9a$$

$$9a - 3a = 2$$

$$6a = 2$$

$$a = \frac{1}{3}$$

(ii) Let $g(x) = \frac{x^2+2}{bx^2+x+4}$.

Given that $\lim_{x \to 1} 2g(x) = \lim_{x \to 0} g(x)$, find the value of b. [5]

$$\lim_{x \to 1} 2g(x) = \lim_{x \to 1} \left(2 \cdot \frac{x^2+2}{bx^2+x+4}\right)$$

$$= \lim_{x \to 1} \left(\frac{2x^2+4}{bx^2+x+4}\right)$$

$$= \frac{2(1)^2+4}{b(1)^2+(1)+4}$$

$$= \frac{2+4}{b+1+4}$$

$$= \frac{6}{b+5}$$

Now,

$$\lim_{x\to 0} g(x) = \lim_{x\to 0}\left(\frac{x^2+2}{bx^2+x+4}\right)$$

$$= \frac{(0)^2+2}{b(0)^2+(0)+4}$$

$$= \frac{2}{4}$$

$$= \frac{1}{2}$$

We are given that:

$$\lim_{x\to 1} 2g(x) = \lim_{x\to 0} g(x)$$

$$\frac{6}{b+5} = \frac{1}{2}$$

$$b + 5 = 12$$

$$b = 12 - 5$$

$$b = 7$$

$\therefore$ The value of $b = 7$.

(b) (i) Let $y = \frac{1}{\sqrt{x}}$. Using first principles, find $\frac{dy}{dx}$. [8]

$$y = \frac{1}{\sqrt{x}}$$

Using first principles,

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{\frac{1}{\sqrt{x+h}} - \frac{1}{\sqrt{x}}}{h}$$

$$= \lim_{h \to 0} \frac{\sqrt{x} - \sqrt{x+h}}{\sqrt{x+h}\sqrt{x}} \div h$$

$$= \lim_{h \to 0} \left(\frac{\sqrt{x} - \sqrt{x+h}}{\sqrt{x+h}\sqrt{x}} \cdot \frac{\sqrt{x} + \sqrt{x+h}}{\sqrt{x} + \sqrt{x+h}} \right) \times \frac{1}{h}$$

$$= \lim_{h \to 0} \left(\frac{x - (x+h)}{(\sqrt{x+h}\sqrt{x})(\sqrt{x} + \sqrt{x+h})} \right) \times \frac{1}{h}$$

$$= \lim_{h \to 0} \frac{-h}{(\sqrt{x+h}\sqrt{x})(\sqrt{x} + \sqrt{x+h})} \times \frac{1}{h}$$

$$= \lim_{h \to 0} \frac{-1}{(\sqrt{x+h}\sqrt{x})(\sqrt{x} + \sqrt{x+h})}$$

$$= \frac{-1}{(\sqrt{x}\sqrt{x})(\sqrt{x} + \sqrt{x})}$$

$$= \frac{-1}{x(2\sqrt{x})}$$

$$= \frac{-1}{2x^{\frac{3}{2}}}$$

$$= -\frac{1}{2} x^{-\frac{3}{2}}$$

(ii) If $y = \frac{x}{\sqrt{1+x}}$, determine an expression for $\frac{dy}{dx}$.

Simplify the answer FULLY. [4]

$$y = \frac{x}{\sqrt{1+x}}$$

$$y = \frac{x}{(1+x)^{\frac{1}{2}}}$$

which is in the form $y = \frac{u}{v}$.

Let $u = x$, $v = (1+x)^{\frac{1}{2}}$

$\frac{du}{dx} = 1$ $\frac{dv}{dx} = \frac{1}{2}(1+x)^{-\frac{1}{2}}$

Using the quotient rule,

$$\frac{dy}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

$$= \frac{(1+x)^{\frac{1}{2}}(1) - (x)\left(\frac{1}{2}\right)(1+x)^{-\frac{1}{2}}}{\left[(1+x)^{\frac{1}{2}}\right]^2}$$

$$= \frac{(1+x)^{\frac{1}{2}} - \frac{1}{2}x(1+x)^{-\frac{1}{2}}}{(1+x)}$$

$$= \frac{\frac{1}{2}(1+x)^{-\frac{1}{2}}[2(1+x)-x]}{(1+x)}$$

$$= \frac{2+2x-x}{2(1+x)(1+x)^{\frac{1}{2}}}$$

$$= \frac{x+2}{2(1+x)^{\frac{3}{2}}}$$

(c) The parametric equations of a curve are given by

$$x = \cos\theta, \qquad y = \sin\theta, \qquad 0 \le \theta \le 2\pi.$$

Find $\frac{dy}{dx}$ in terms of θ.

Simplify the answer as far as possible. [4]

$x = \cos\theta$, $y = \sin\theta$

$\frac{dx}{d\theta} = -\sin\theta$ $\frac{dy}{d\theta} = \cos\theta$

Using the chain rule,

$$\frac{dy}{dx} = \frac{dy}{d\theta} \times \frac{d\theta}{dx}$$

$$= \cos\theta \times \left(\frac{1}{-\sin\theta}\right)$$

$$= -\frac{\cos\theta}{\sin\theta}$$

$$= -\cot\theta$$

Total: 25 marks

June 2014 – Question 6 – Solution

(a) The gradient of a curve which passes through the point $(-1, -4)$ is given by

$$\frac{dy}{dx} = 3x^2 - 4x + 1$$

(i) Find

(a) the equation of the curve [4]

$$\frac{dy}{dx} = 3x^2 - 4x + 1$$

$$y = \int \frac{dy}{dx}\, dx$$

$$y = \int (3x^2 - 4x + 1)\, dx$$

$$y = \frac{3x^3}{3} - \frac{4x^2}{2} + x + c$$

$$y = x^3 - 2x^2 + x + c \qquad \text{(G.S.)}$$

Substituting $(-1, -4)$ into the G.S. gives:

$$-4 = (-1)^3 - 2(-1)^2 + (-1) + c$$

$$-4 = -1 - 2 - 1 + c$$

$$-4 = -4 + c$$

$$c = -4 + 4$$

$$c = 0$$

$\therefore$ The equation of the curve is: $y = x^3 - 2x^2 + x$.

(b) the coordinates of the stationary points and determine their nature. [8]

$\frac{dy}{dx} = 3x^2 - 4x + 1$

At stationary points, $\frac{dy}{dx} = 0$.

When $\frac{dy}{dx} = 0$,

$3x^2 - 4x + 1 = 0$

$(3x - 1)(x - 1) = 0$

Either $3x - 1 = 0$ or $x - 1 = 0$

$x = \frac{1}{3}$ $x = 1$

When $x = \frac{1}{3}$,

$y = \left(\frac{1}{3}\right)^3 - 2\left(\frac{1}{3}\right)^2 + \left(\frac{1}{3}\right)$

$= \frac{1}{27} - 2\left(\frac{1}{9}\right) + \frac{1}{3}$

$= \frac{1}{27} - \frac{2}{9} + \frac{1}{3}$

$= \frac{4}{27}$

When $x = 1$,

$y = (1)^3 - 2(1)^2 + (1)$

$= 1 - 2 + 1$

$= 0$

Now,

$$\frac{d^2y}{dx^2} = 6x - 4$$

When $x = \frac{1}{3}$,

$$\frac{d^2y}{dx^2} = 6\left(\frac{1}{3}\right) - 4$$

$$= 2 - 4$$

$$= -2 \qquad (< 0)$$

Since $\frac{d^2y}{dx^2} < 0$, then $\left(\frac{1}{3}, \frac{4}{27}\right)$ is a maximum point.

When $x = 1$,

$$\frac{d^2y}{dx^2} = 6(1) - 4$$

$$= 6 - 4$$

$$= 2 \qquad (> 0)$$

Since $\frac{d^2y}{dx^2} > 0$, then $(1, 0)$ is a minimum point.

(ii) Sketch the curve in (a)(i)(a) above, clearly marking ALL stationary points and intercepts. [4]

$y = x^3 - 2x^2 + x$

When $y = 0$,

$x^3 - 2x^2 + x = 0$

$x(x^2 - 2x + 1) = 0$

$x(x-1)(x-1) = 0$

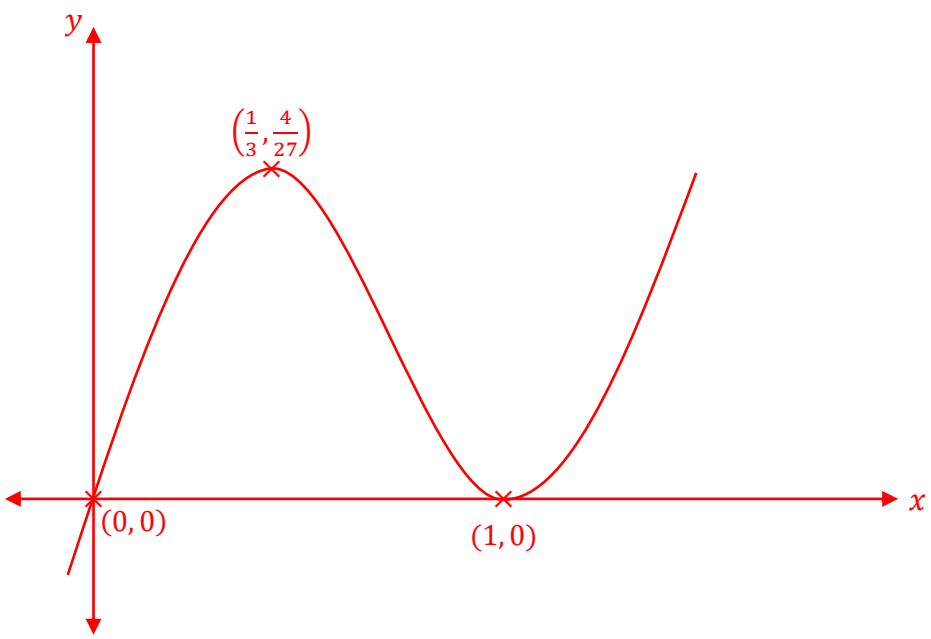
Either $x = 0$ or $x - 1 = 0$

$x = 1$

$\therefore$ The curve cuts the x-axis at $(0, 0)$ and $(1, 0)$.

$\left(\frac{1}{3}, \frac{4}{27}\right)$ is a maximum point and $(1, 0)$ is a minimum point.

The sketch is as follows:



(b) The equation of a curve is given by $f(x) = 2x\sqrt{1+x^2}$.

(i) Evaluate $\int_0^3 f(x)\,dx$. [5]

$f(x) = 2x\sqrt{1+x^2}$

Let $t = 1 + x^2$

$\frac{dt}{dx} = 2x$

$dx = \frac{dt}{2x}$

When $x = 0$,

$t = 1 + (0)^2$

$= 1 + 0$

$= 1$

When $x = 3$,

$t = 1 + (3)^2$

$= 1 + 9$

$= 10$

Now,

$\int_0^3 f(x)\,dx = \int_0^3 2x\sqrt{1+x^2}\ \ dx$

$= \int_1^{10} 2x \cdot \sqrt{t}\ \frac{dt}{2x}$

$= \int_1^{10} t^{\frac{1}{2}}\ dt$

$= \left[\frac{t^{\frac{3}{2}}}{\frac{3}{2}}\right]_1^{10}$

$= \frac{2}{3}\left[t^{\frac{3}{2}}\right]_1^{10}$

$= \frac{2}{3}\left[(10)^{\frac{3}{2}} - (1)^{\frac{3}{2}}\right]$

$= \frac{2}{3}\left(\sqrt{1000} - 1\right)$

(ii) Find the volume generated by rotating the area bounded by the curve in (b)(i) above, the x-axis, and the lines $x = 0$ and $x = 2$ about the x-axis. [4]

$$y = 2x\sqrt{1 + x^2}$$

$$y = 2x(1 + x^2)^{\frac{1}{2}}$$

$$y^2 = 4x^2(1 + x^2)$$

$$y^2 = 4x^2 + 4x^4$$

Now,

$$\text{Volume} = \pi \int_a^b y^2\, dx$$

$$= \pi \int_0^2 (4x^2 + 4x^4)\, dx$$

$$= \pi \left[\frac{4x^3}{3} + \frac{4x^5}{5}\right]_1^2$$

$$= \pi \left[\left(\frac{4(2)^3}{3} + \frac{4(2)^5}{5}\right) - \left(\frac{4(1)^3}{3} + \frac{4(1)^5}{5}\right)\right]$$

$$= \pi \left[\left(\frac{32}{3} + \frac{128}{5}\right) - \left(\frac{4}{3} + \frac{4}{5}\right)\right]$$

$$= \pi \left(\frac{32}{3} + \frac{128}{5} - 0\right)$$

$$= \frac{544}{15}\pi \text{ units}^3$$

Total: 25 marks